



Figure 1: Google shows various parts of a website, retrieved with semantic information



Figure 2: Movie ratings retrieved with semantic information

```
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/"
  xmlns:property="http://www.example.org/properties/">
<br>
  <rdf:Description rdf:about="http://www.linuxforu.com/semanticweb.html">
    <property:author>Ankit Mathur</property:author>
    <dc:language>en</dc:language>
    <dc:title>Semantic Web</dc:title>
  </rdf:Description>
</rdf:RDF>
```

Another possible notation, in RDF-Turtle, is as follows:

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix dc: <http://purl.org/dc/elements/1.1/> .
@prefix ex: <http://example.org/properties/> .

< http://www.linuxforu.com/semanticweb.html >
  Property:author "Ankit Mathur";
  dc:title "Semantic Web";
  dc:language "en";
  ] .
```

You can see that it's easy to represent information about an article in many notations, though RDF/XML is a little more verbose if we compare them. But why did RDF come into the picture, when we could have done the same with XML and schemas? Well, for one, XML is too flexible and schemas are too restrictive. When using XML schemas, we have to define the order and the possible values contained within the elements. Also, RDF can be used for annotation with documents, and is easily extensible, which is simply not possible with other technologies.

Resource Description Framework Schema

RDFS is basically used to describe properties and classes in RDF, so it is an extension that builds upon RDF itself. Here, I will quote the ever-popular example of cats and animals, where we can describe a Tiger, which is a type of animal, etc.

This can be expressed in RDFS as follows:

```
<?xml version="1.0"?>
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
  xml:base="http://www.example.com/animals#">
  <rdfs:Class rdf:ID="animal" />
  <rdfs:Class rdf:ID="tiger">
    <rdfs:subClassOf rdf:resource="#animal"/>
  </rdfs:Class>
</rdf:RDF>
```

Ontologies: Web Ontology Language

Ontologies have often been talked about in computer science books. OWL, or Web Ontology Language, deals with the representation of these ontologies. But what exactly are ontologies? These are used to describe a domain of information and develop relationships between them—for example, classes, properties and their relationships. These ontologies are used to describe and develop a semantic for each of the resources available on the Web, from which machine-processable information can be derived. This is just an enhanced form of what RDFS provides, and has more features and complexity. The OWL family consists of more than one sub-language and syntax. The sub-languages are:

- OWL Lite
- OWL DL
- OWL Full

To get a detailed overview of what OWL entails, you can pick up a book on ontologies and go through it first.

SPARQL

SPARQL is a query language used to retrieve information from the Semantic Web stack and its RDF representation. It is more verbose than SQL, and allows for more complex queries. There are various types of queries, depending on the usage:

- SELECT query
- DESCRIBE query
- ASK query